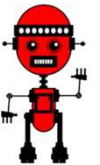


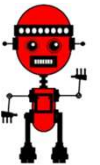
PYTHON FLASK TUTORIAL



CONTENT



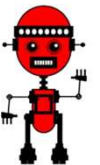
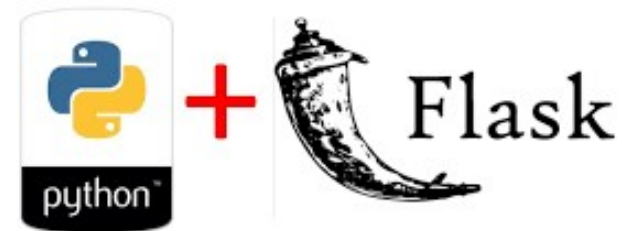
- Overview
- Creation of virtual environment
- basic app
- Jinja2 template
- App routing
- URL Building
- HTTP method, HTTP templates
- Request Object



Overview

- Flask is a web framework that provides libraries to build lightweight web applications in python.
- Developed by **Armin Ronacher** who leads an international group of python enthusiasts (POCCO)
- Based on WSGI toolkit and jinja2 template engine.
- Flask is considered as a micro framework.

REST API using Flask



Creation of virtual environment

virtualenv is considered as the virtual python environment builder which is used to create the multiple python virtual environment side by side.

It is always advisable to create virtual environment to manage the dependencies for your project, both in development and in production.

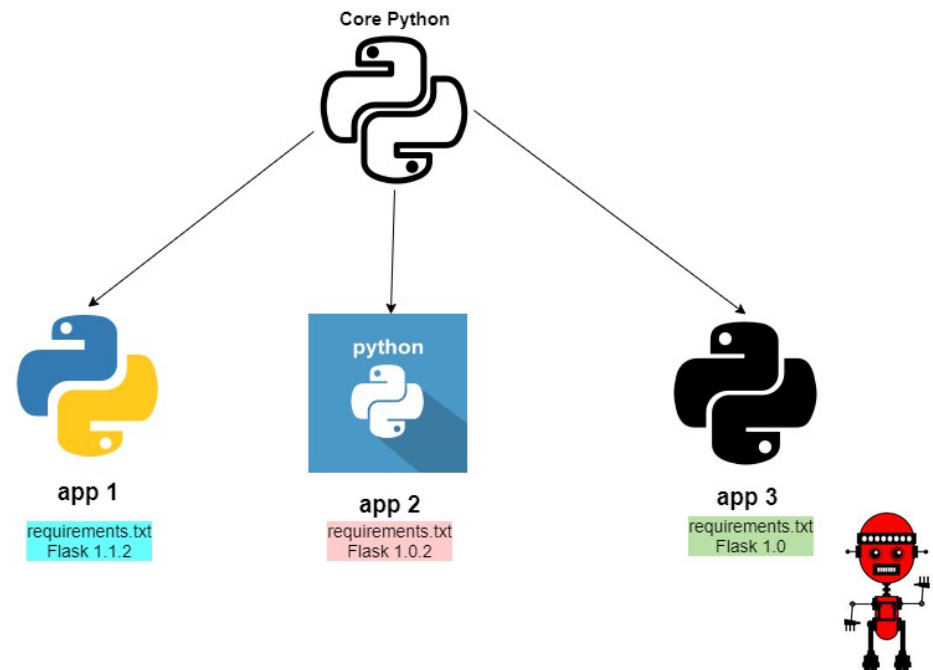
```
>> pip install virtualenv
```

```
>> virtualenv name_of_virtualenv
```

```
>> cd <envname>
```

```
>> Scripts\activate
```

```
>> pip install flask
```



Basic App

```
from flask import Flask

app = Flask(__name__)          #creating the Flask class object

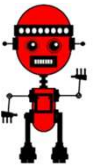
@app.route('/')    #decorator

def home():

    return "hello, this is our first flask website";

if __name__ == '__main__':

    app.run(debug = True)
```



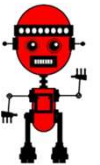
Route() method

The route() function of the Flask class defines the URL mapping of the associated function.

```
app.route(rule, options)
```

rule: It represents the URL binding with the function.

options: It represents the list of parameters to be associated with the rule object

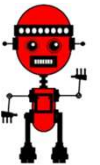


Run() method

run method of the Flask class is used to run the flask application on the local development server

```
app.run(host, port, debug, options)
```

S N	Option	Description
1	host	The default hostname is 127.0.0.1, i.e. localhost.
2	port	The port number to which the server is listening to. The default port number is 5000.
3	debug	The default is false. It provides debug information if it is set to true.
4	options	It contains the information to be forwarded to the server.

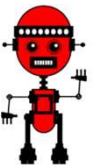


url binding

url_for() function is used to build a URL to the specific function dynamically

avoids hard coding of the URLs.

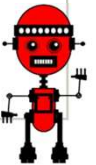
```
@app.route('/user/<name>')  
def hello_user(name):  
    if name == 'admin':  
        return redirect(url_for('hello_admin'))
```



Flask HTTP methods

- HTTP is the hypertext transfer protocol -- the foundation of the data transfer in the world wide web
- All web frameworks including flask need to provide several HTTP methods for data communication

Method	Description
GET	It is the most common method which can be used to send data in the unencrypted form to the server.
HEAD	It is similar to the GET but used without the response body.
POST	It is used to send the form data to the server. The server does not cache the data transmitted using the post method.
PUT	It is used to replace all the current representation of the target resource with the uploaded content.
DELETE	It is used to delete all the current representation of the target resource specified in the URL.



Flask Templates

flask facilitates us to return the response in the form of HTML templates

```
from flask import *
app = Flask(__name__)

@app.route('/')
def message():
    return "<html><body><h1>Hi,
welcome to the
website</h1></body></html>"

if __name__ == '__main__':
    app.run(debug = True)
```

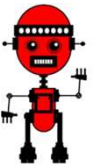
Rendering external HTML files

Flask provides us the `render_template()` function which can be used to render the external HTML file to be returned as the response from the view function.

```
from flask import *
app = Flask(__name__)

@app.route('/')
def message():
    return render_template('message.html')

if __name__ == '__main__':
    app.run(debug = True)
```



Jinja2 template

Python library that allows us to build expressive and extensible templates. It has special placeholders to serve dynamic data. A Jinja template file is a text file that does not have a particular extension.

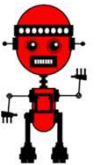
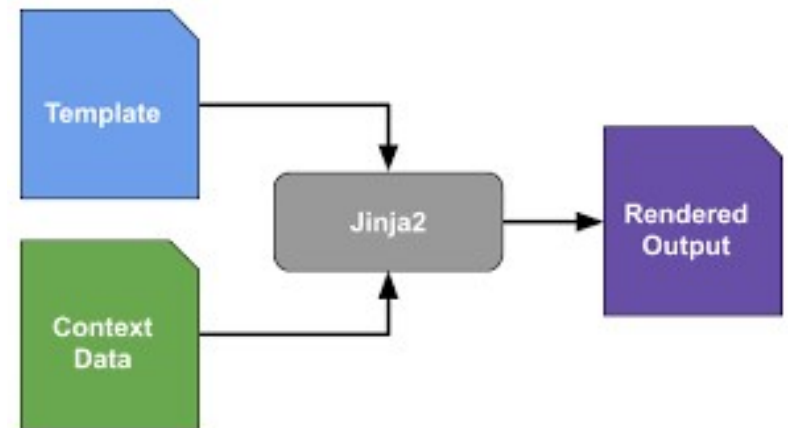
The jinja2 template engine provides the following delimiters to escape from the HTML.

`{% ... %}` for statements

`{{ ... }}` for expressions to print to the template output

`{# ... #}` for the comments that are not included in the template output

`# ... ##` for line statements



Flask Request Object

The data from a client's web page is sent to the server as a global request object.

Important attributes of request object are listed below –

Form – It is a dictionary object containing key and value pairs of form parameters and their values.

args – parsed contents of query string which is part of URL after question mark (?).

Cookies – dictionary object holding Cookie names and values.

files – data pertaining to uploaded file.

method – current request method.

